

Finite-Fringe Rigidity in Congruence-Restricted Greedy Dissociated Sequences

Proofs, exact formulas, and inverse transseries for OEIS A139217 and A139218

Merged from two independent manuscripts (batch 73O1, sources 50 and 43)

ProveIt research report (AI-assisted)
prepared for Vladimir Reshetnikov with OpenAI

1 October 2026

Abstract

A finite set of positive integers is *dissociated* when all of its subset sums are distinct. OEIS A139217 and A139218 are obtained by a greedy rule: at each step choose the least admissible positive integer congruent to 1 or 2 modulo 3, respectively. Both entries conjecture an eventual period-three law for $a(n+1) - 2a(n)$ and the same order-three linear recurrence.

We prove substantially stronger statements. After three terms, the signed subset-sum difference set is a complete interval with a fixed finite fringe of holes. For A139217 the only positive fringe hole is $S_n - 3$; for A139218 the positive fringe holes are $S_n - h$ with $h \in \{1, 3, 6, 11\}$, where $S_n = \sum_{j \leq n} a(j)$. A general finite-fringe propagation theorem shows that these descriptions persist forever. The greedy choice then becomes a finite residue calculation, giving exact period-three corrections, rational generating functions, closed forms, asymptotics, logarithmic transseries, and residue-branch inverse transseries.

In particular, for A139217,

$$a(n) = \frac{3}{4} 2^n + \begin{cases} 1, & n \equiv 0, 1 \pmod{3}, \\ -2, & n \equiv 2 \pmod{3}, \end{cases} \quad (n \geq 3),$$

and for A139218,

$$a(n) = \frac{39}{56} 2^n + \begin{cases} -25/7, & n \equiv 0 \pmod{3}, \\ 20/7, & n \equiv 1 \pmod{3}, \\ 5/7, & n \equiv 2 \pmod{3}, \end{cases} \quad (n \geq 4).$$

Both satisfy $a(n) = a(n-1) + a(n-2) + 2a(n-3)$ on their correct eventual ranges. For A139217 that range is $n \geq 6$; the current OEIS statement “ $n > 4$ ” is false at $n = 5$. Self-contained exact verifiers are supplied. The proofs are conventional mathematics, not a formal Lean development, and independent review is recommended before OEIS submission.

This report merges two manuscripts that proved these theorems independently on the same day: source 50 (the base text, with the general finite-fringe propagation theorem) and source 43 (boundary-defect coordinates). Each shared theorem is printed once and credited to both sources; source 43’s different proofs are kept as marked second routes, and its additional results are printed in full: exact cross-identities between the two sequences, the sizes of the signed difference sets, an explicit remainder bound and explicit coefficients for the inverse series, counting asymptotics, and tables of terms and defects for $n \leq 14$.

Contents

1	Introduction	3
2	Dissociation, signed differences, and ternary supports	7
3	The finite-fringe propagation mechanism	8
4	OEIS A139217: the residue-one case modulo 3	13
5	OEIS A139218: the residue-two case modulo 3	15
6	Relations between the two sequences (source 43)	18
7	Exact asymptotics and transseries	19
8	Generating functions and recurrence diagnostics	22
9	Exact computational audit	23
10	Proposed OEIS updates	24
11	Formalization roadmap for ProveIt	25
12	Further research questions	27
13	Conclusion	30
A	Seed support certificates	31
B	Derivation of the rational generating functions	31
C	Audit checklist	32
D	Tables of terms, partial sums, and defects (source 43)	32
E	Provenance and merge record	33

1 Introduction

1.1 The two OEIS problems

Let $A_n = \{a(1), \dots, a(n)\}$. The defining condition in the two entries is that every two subsets of A_n have different sums. In modern additive terminology, A_n is a *dissociated set*. The greedy rule is constrained to one arithmetic progression:

A139217: $a(n+1)$ is the least positive $b \equiv 1 \pmod{3}$ for which $A_n \cup \{b\}$ is dissociated,

A139218: $a(n+1)$ is the least positive $b \equiv 2 \pmod{3}$ for which $A_n \cup \{b\}$ is dissociated.

The entries currently list

A139217: 1, 4, 7, 13, 22, 49, 97, 190, 385, 769, 1534, 3073, 6145, 12286, $\dots$,

A139218: 2, 5, 8, 14, 23, 41, 92, 179, 353, 716, 1427, 2849, 5708, 11411, $\dots$

Both entries conjecture that the deviations $a(n+1) - 2a(n)$ are eventually periodic and that

$$a(n) = a(n-1) + a(n-2) + 2a(n-3) \quad (1.1)$$

eventually. See OEIS A139217 and OEIS A139218 [7, 8], accessed 1 October 2026. The entries were contributed by John W. Layman in April 2008. Both also mention the analogous odd-integer construction, which produces a shifted Jacobsthal sequence. As of the access date the order-three formulas are labelled “It appears”, with stated ranges $n > 4$ and $n > 6$ respectively (source 43).

The aim of this report is not merely to verify more terms. We identify a stable structural invariant of the complete signed difference set. Once that invariant is found, the conjectured recurrences become short consequences of a finite residue table.

1.2 Main results

Write

$$S_n := \sum_{j=1}^n a(j), \quad \mathcal{D}_n := \left\{ \sum_{j=1}^n \varepsilon_j a(j) : \varepsilon_j \in \{-1, 0, 1\} \right\}.$$

The strongest concrete conclusions are as follows. Both theorems were proved independently by sources 50 and 43 (source 43’s Theorems 4.1 and 5.1).

Theorem 1.1 (A139217, full signed support; sources 50 and 43). *For every $n \geq 3$,*

$$\mathcal{D}_n = [-S_n, S_n] \cap \mathbb{Z} \setminus \{-(S_n - 3), S_n - 3\}.$$

Consequently,

$$a(n+1) - 2a(n) = \begin{cases} -1, & n \equiv 0 \pmod{3}, \\ -4, & n \equiv 1 \pmod{3}, \\ 5, & n \equiv 2 \pmod{3}, \end{cases} \quad (n \geq 3),$$

and the recurrence (1.1) holds exactly for $n \geq 6$.

Theorem 1.2 (A139218, full signed support; sources 50 and 43). *For every $n \geq 3$,*

$$\mathcal{D}_n = [-S_n, S_n] \cap \mathbb{Z} \setminus \{\pm(S_n - h) : h \in \{1, 3, 6, 11\}\}.$$

Consequently,

$$a(n+1) - 2a(n) = \begin{cases} 10, & n \equiv 0 \pmod{3}, \\ -5, & n \equiv 1 \pmod{3}, \\ -5, & n \equiv 2 \pmod{3}, \end{cases} \quad (n \geq 4),$$

and the recurrence (1.1) holds exactly for $n \geq 7$.

The full interval-minus-fringe formulas are stronger than the OEIS conjectures: they determine every signed relation target, not only the next greedy term.

Source 43 records the resolved OEIS statements as follows.

- (1) For A139217, (1.1) holds for every $n \geq 6$. The OEIS range $n > 4$ must be corrected: at $n = 5$, $22 \neq 13 + 7 + 2 \cdot 4 = 28$.
- (2) For A139218, (1.1) holds for every $n \geq 7$, exactly matching the conjectured range $n > 6$.
- (3) Both sequences have exact dyadic-plus-periodic closed forms, rational ordinary generating functions, and exact logarithmic staircase inverses.
- (4) The recurrence is a corollary of a stronger signed-difference-set classification with finitely many persistent boundary defects.

The main proofs are elementary once the correct invariant is found. The nontrivial step is not solving a recurrence guessed from data, but proving that the greedy construction continues to choose exactly the terms predicted by that recurrence. The signed difference set provides the missing certificate (source 43).

1.3 Relation to recent work

Dissociated sets have a long history in distinct-subset-sum problems; useful classical references include Cantor–Mills [3] and Lunnon [6], while modern additive terminology is discussed by Candela–Helfgott [2] and Tao–Vu [12]. The general distinct-subset-sums problem is a classical topic in additive combinatorics; see, for example, Bohman’s construction [1], the lower-bound work of Dubroff–Fox–Xu [4], and Steinerberger’s analytic reformulation [11] (source 43). A recent paper of Dutta [5] studies the unrestricted greedy algorithm and proves that, from any two fixed initial terms, the unrestricted greedy sequence eventually doubles exactly.

The arithmetic-progression restriction changes the phenomenon. Exact doubling usually leaves the permitted residue class: if $a(n) \equiv 1 \pmod{3}$, then $2a(n) \equiv 2 \pmod{3}$, and conversely. The correct replacement is *doubling plus a bounded periodic correction*. The finite-fringe theorem below explains why; its threshold form (Corollary 3.4) is the explicit constrained analogue of Dutta’s doubling (source 43).

A targeted search on 1 October 2026 found the conjectures in the OEIS entries and the general recent paper [5], but did not locate a published proof of these two exact progression-restricted formulas. This is a research report, not a claim of priority after an exhaustive literature review. Source 43’s independent search used the exact OEIS identifiers, initial terms and recurrence and likewise found no published proof; because such a search can miss offline, unpublished or poorly indexed sources, its priority statement is deliberately modest too.

1.4 Status and reproducibility

All arguments are written as ordinary proofs. The accompanying verifier `code/50-finite-fringe-verify.py` performs exact integer checks of the greedy construction, the fringe identities through 22 terms, the published prefixes, the closed forms, the recurrences, and the rational generating functions. The second verifier `code/43-boundary-verify.py` (source 43) works with the signed difference sets directly and also checks the counting functions, the cross-identities and the failure at $n = 5$ (Section 9). Computation is used as an audit, not as a substitute for induction. No theorem in this report has been formalized in Lean at the time of writing.

Source 43 states its status as follows. It is an unrefereed research draft. The proofs are written to be independently checkable and are accompanied by exact verification code, but they

have not been formalized in Lean or independently peer reviewed. The literature and priority search was targeted rather than exhaustive; no claim of global priority is made beyond the precise statements and sources recorded here. In particular source 43 does *not* claim:

- that no equivalent proof exists elsewhere;
- that the general congruence-restricted greedy problem has been classified;
- that the results have passed peer review or formal verification;
- that exact computations alone establish the infinite claims.

The accompanying programs check finite prefixes and algebraic identities, while the inductions in this report prove the statements for all indices.

The ProveIt repository was inspected at commit `c79d64038558` [9] (source 50). The present topic is separate from its existing A290268 development, but the report follows the same useful pattern: an exact support model, a structural theorem, executable checks, and an explicit formalization roadmap.

[Merge note, batch 73O1.] The A290268 development meant here is the collection report `a290268-unbounded-deficits` of batch 72A; there is no mathematical link between it and this report.

The ProveIt repository combines formal mathematics, executable certificates, and reproducible research artifacts, with a clear separation between exploratory computation and trusted proof boundaries; the human-readable proof is primary, and the verifiers are reproducibility aids (source 43). The inverse expansions also fit the repository’s broader series-and-transseries programme [10], although the expansions obtained here are unusually simple: on each residue branch they are convergent Taylor series in $1/x$ (source 43).

1.5 The two sources and this merge

This report is built from two manuscripts of batch 73O1 of ProveIt’s incoming-reports intake, written independently on 1 October 2026.

- **Source 50** (the base text), *Finite-Fringe Rigidity in Congruence-Restricted Greedy Dissociated Sequences*: the ternary support $\mathcal{E}(A)$, the general finite-fringe propagation theorem with its collision condition, the residue rule for an arbitrary modulus, the periodic correction principle, logarithmic and ratio transseries, ternary seed certificates and ten research questions. Its section order is kept.
- **Source 43**, *Boundary-Defect Rigidity in Congruence-Restricted Greedy Dissociated Sets*: the same fringe theorems, closed forms, recurrence ranges, generating functions, inverse branches and counting functions, proved in signed (boundary-defect) coordinates.

Every result, proof, example, remark, question and non-claim of both manuscripts is printed. A theorem proved by both is printed once, in source 50’s wording, and credited “sources 50 and 43”. Where source 43’s proof is genuinely different, it follows as a “Second proof (source 43)”. Source 43’s propagation theorem is a corollary of source 50’s (Corollary 3.4); its periodic forcing principle is source 50’s periodic correction principle restated for partial sums (Proposition 3.7). Material only source 43 has is marked (source 43): Section 6, the forward asymptotics and spectral form in Section 7.1, the remainder bound and explicit coefficients in Section 7.3, the counting asymptotics, the second verifier, its proposed OEIS wording, its roadmap, its research questions, and Appendix D. Text added in the merge is marked **[Merge note, batch 73O1.]**. Appendix E records the provenance and every choice the merge made.

1.6 Notation across the two sources

The two manuscripts use different letters for the same notions, and two letters with different meanings. This report uses source 50’s notation throughout; source 43’s material is translated as in the table.

Notion	This report	Source 50	Source 43
terms	$a(n); u_n, v_n$	$a(n)$	u_n (A139217), v_n (A139218), a_n
partial sums	$S_n; U_n, V_n$	S_n	U_n, V_n, S_n
signed difference set	$\mathcal{D}(A), \mathcal{D}_n$	$\mathcal{D}$	$\Delta(A), \Delta_n^{(u)}, \Delta_n^{(v)}$
ternary support	$\mathcal{E}(A)$	$\mathcal{E}$	—
fringe offsets, max	$H, M = \max H$	H, M	C, M (renamed)
interval with fringe	$\mathcal{F}_H(S)$	(3.2)	$\mathcal{F}_C(S)$
greedy correction	$c_n = a(n+1) - S_n$	c_n	$-d_n$ (A139217), $-e_n$ (A139218) (renamed, sign)
doubling deviation	$d_n = a(n+1) - 2a(n)$	d_n	no symbol
leading constant	C	C	α (renamed)
periodic part of $a(n)$	$p_n, p_j; r_n$ for v_n	p_n, p_j	p_n (u), r_n (v), p
periodic part of S_n	q_n (U), s_n (V)	—	q_n, s_n
floor-log count	Λ	Λ	Φ (renamed)
inverse truncation order	K	—	M (renamed)
forward shift operator	τ	—	E (renamed)
generating-function	x	x	z (renamed)
variable			

Readings to avoid.

- Source 43’s d_n is the *offset* $S_n - a(n+1)$ of the greedy term below the partial sum, so $d_n^{(43)} = -c_n$. It is *not* source 50’s deviation $d_n = a(n+1) - 2a(n)$. For A139217, source 43’s $d_n = (-1, 3, -2)$ is $-c_n$ with $c_n = (1, -3, 2)$, while the deviation is $d_n = (-1, -4, 5)$.
- Source 43’s C is a finite *set* of offsets (this report’s H); source 50’s C , used here, is the leading *constant* of $a(n) = C2^n + p_n$.
- Source 43’s M is used both for $\max C$ and for a truncation order; here $M = \max H$ only, and the truncation order is K .
- $\mathcal{E}(A)$ is the ternary support; source 43’s shift operator E is written τ .
- The letter p is source 50’s period $p = m/\gcd(m, r)$ (Section 3.4), while p_n and p_j denote periodic parts; a subscript always distinguishes them.
- The word “transseries” in source 50’s title and Section 7 is used loosely: every expansion in this report is a convergent series in 2^{-n} or $1/x$, with no exponentially small sectors or trans-monomials.

2 Dissociation, signed differences, and ternary supports

2.1 Equivalent forms of dissociation

For a finite set $A \subset \mathbb{N}$, define its ordinary subset-sum set and its signed difference set by

$$\Sigma(A) := \left\{ \sum_{a \in B} a : B \subseteq A \right\}, \quad (2.1)$$

$$\mathcal{D}(A) := \left\{ \sum_{a \in A} \varepsilon_a a : \varepsilon_a \in \{-1, 0, 1\} \right\}. \quad (2.2)$$

Also write $S(A) = \sum_{a \in A} a$.

Lemma 2.1 (standard equivalences). *For a finite set A of positive integers, the following are equivalent.*

- (i) *All subset sums of A are distinct.*
- (ii) *The only relation $\sum_{a \in A} \varepsilon_a a = 0$ with $\varepsilon_a \in \{-1, 0, 1\}$ is the trivial relation.*
- (iii) *If $b > 0$, then $A \cup \{b\}$ is dissociated exactly when $b \notin \mathcal{D}(A)$, provided A is dissociated.*

Proof. An equality between two subset sums becomes a signed relation after canceling the common elements, and every signed relation can be separated into its positive and negative supports. This proves (i) $\Leftrightarrow$ (ii). For (iii), a relation involving the new element has coefficient $+1$ or -1 on b , hence is equivalent to $b \in \mathcal{D}(A)$. A relation with coefficient zero on b would already contradict dissociation of A . $\square$

Statement (iii) is source 43's *append criterion* (its Lemma 2.2), which it calls decisive for the greedy algorithm.

Second proof of (iii) (source 43). The subset sums of $A \cup \{b\}$ split into two copies, $\Sigma(A)$ and $b + \Sigma(A)$. Each copy has no internal collision because A is dissociated. A collision between the copies is an equality $b + s = t$ with $s, t \in \Sigma(A)$, equivalently $b = t - s \in \Sigma(A) - \Sigma(A) = \mathcal{D}(A)$. Conversely, every representation $b = t - s$ creates that collision. $\square$

The equality

$$\mathcal{D}(A) = \Sigma(A) - \Sigma(A)$$

is immediate. In particular, $\mathcal{D}(A)$ is symmetric and lies in $[-S(A), S(A)]$. Thus the greedy rule is not mysterious: at stage n , all positive elements of $\mathcal{D}(A_n)$ are forbidden, and the next term is the least admissible integer outside that set (source 43).

Remark 2.2 (source 43). Dutta uses the same signed-difference-set criterion in the unrestricted greedy setting [5]. The new ingredient here is an exact description of the whole support of $\mathcal{D}(A_n)$ for the two congruence-restricted OEIS sequences.

2.2 The shifted ternary support

The central device is the support obtained by allowing digits 0, 1, 2:

$$\mathcal{E}(A) := \left\{ \sum_{a \in A} \delta_a a : \delta_a \in \{0, 1, 2\} \right\}. \quad (2.3)$$

Lemma 2.3 (translation identity). *For every finite $A \subset \mathbb{N}$,*

$$\mathcal{E}(A) = S(A) + \mathcal{D}(A).$$

Consequently, $\mathcal{E}(A) \subseteq [0, 2S(A)]$, and $x \in \mathcal{E}(A)$ if and only if $2S(A) - x \in \mathcal{E}(A)$.

Proof. Set $\delta_a = 1 + \varepsilon_a$. Then $\sum \delta_a a = S(A) + \sum \varepsilon_a a$. Replacing every digit δ_a by $2 - \delta_a$ gives the reflection symmetry. $\square$

Equivalently, $\mathcal{E}(A)$ is the exponent support of

$$P_A(x) := \prod_{a \in A} (1 + x^a + x^{2a}). \quad (2.4)$$

Only the support matters here; coefficients may exceed one.

If b is appended, the support has the exact three-copy recursion

$$\mathcal{E}(A \cup \{b\}) = \mathcal{E}(A) \cup (b + \mathcal{E}(A)) \cup (2b + \mathcal{E}(A)). \quad (2.5)$$

This elementary identity drives the entire proof. In signed coordinates it reads $\mathcal{D}(A \cup \{b\}) = \mathcal{D}(A) + \{-b, 0, b\}$, the form source 43 uses.

3 The finite-fringe propagation mechanism

3.1 Fringe-complete supports

For a finite set $H \subset \mathbb{N}$ and an integer T , write $T - H := \{T - h : h \in H\}$ and $H + H := \{h + h' : h, h' \in H\}$.

Definition 3.1 (finite-fringe completeness). Let A have total S . We say that $\mathcal{E}(A)$ has fixed fringe H if

$$\mathcal{E}(A) = ([0, 2S] \cap \mathbb{Z}) \setminus (H \cup (2S - H)). \quad (3.1)$$

We always assume $H \subseteq \{1, \dots, S - 1\}$.

By Lemma 2.3, (3.1) is equivalent to

$$\mathcal{D}(A) = ([-S, S] \cap \mathbb{Z}) \setminus \{\pm(S - h) : h \in H\}. \quad (3.2)$$

Thus the only positive admissible integers at most S are the fringe gaps $S - h$. Source 43 names the right side of (3.2) a *boundary-defect interval*: for H finite and nonempty, $M = \max H$ and $S > M$,

$$\mathcal{F}_H(S) := ([-S, S] \cap \mathbb{Z}) \setminus \{\pm(S - h) : h \in H\}, \quad (3.3)$$

a complete symmetric integer interval with finitely many holes at fixed distances H from its two endpoints.

3.2 The propagation theorem

Theorem 3.2 (finite-fringe propagation). *Let $A \subset \mathbb{N}$ have total S , and suppose (3.1) holds for a nonempty finite set H . Put $M = \max H$. Let b be a positive integer such that*

$$M < b \leq 2S - M, \quad 2S - b \notin H + H. \quad (3.4)$$

Then

$$\mathcal{E}(A \cup \{b\}) = [0, 2(S + b)] \cap \mathbb{Z} \setminus (H \cup (2(S + b) - H)). \quad (3.5)$$

In words, appending b preserves the same finite fringe.

Proof. Put $E = \mathcal{E}(A)$ and consider the three translates in (2.5). Their ambient intervals are

$$I_0 = [0, 2S], \quad I_1 = [b, b + 2S], \quad I_2 = [2b, 2b + 2S].$$

Because $b \leq 2S - M < 2S$, these intervals cover the whole interval $[0, 2S + 2b]$.

First consider a point covered only by I_0 . It is smaller than b . The reflected holes $2S - H$ are at least $2S - M \geq b$, so the only holes that can occur there are the desired left fringe H . By reflection, a point covered only by I_2 is missing exactly when it lies in the desired right fringe $2(S + b) - H$.

A point covered only by I_1 can occur only when $b > S$. Writing the point as $b + y$, exclusivity gives

$$2S - b < y < b.$$

The lower holes satisfy $h \leq M \leq 2S - b$, while the upper holes satisfy $2S - h \geq 2S - M \geq b$. Hence y is not a hole of E , so every point in the middle-only region is covered.

It remains to rule out an uncovered point in an overlap. In an overlap of two consecutive translates, such a point would require two holes of E whose difference is b . Since $b > M$, two lower holes cannot differ by b , and two upper holes cannot differ by b . The only possible alignment is

$$2S - h = b + h' \quad (h, h' \in H),$$

which is equivalent to $2S - b = h + h' \in H + H$. This is excluded by (3.4). Any overlap of I_0 and I_2 is automatically a triple overlap, so a consecutive pair is available there as well.

Finally, $h \in H$ lies below b and hence cannot be filled by a shifted copy; similarly $2(S + b) - h$ lies beyond the end of I_1 and can only come from the missing reflected point in I_2 . Therefore the endpoint holes remain, and all other points are present. $\square$

The collision condition has a transparent meaning: a high hole in one copy must not land on a low hole in the next copy.

Corollary 3.3 (automatic persistence beyond a threshold). *Fix a modulus m and a fringe H with $M = \max H$. Suppose a candidate has the form $b = S + c$ with $-M \leq c \leq m$. If*

$$S > 2M + m,$$

then all conditions of Theorem 3.2 hold.

Proof. We have $b \geq S - M > M$, and $b \leq S + m < 2S - M$. Also $2S - b = S - c \geq S - m > 2M$, whereas $H + H \subseteq [2, 2M]$. $\square$

Thus a fixed fringe, once reached beyond a finite threshold, is rigid. No asymptotic estimate is needed.

Source 43 proves the threshold form of this persistence directly, in signed coordinates and for an arbitrary correction. It is the case of Theorem 3.2 recorded next (source 43's Theorem 3.2, "boundary-defect propagation").

Corollary 3.4 (threshold form; sources 50 and 43). *Let $H \subset \mathbb{N}$ be finite and nonempty, $M = \max H$, and let $c \in \mathbb{Z}$. Suppose*

$$S > 2M + |c|. \tag{3.6}$$

Set

$$b = S + c, \quad S' = S + b = 2S + c.$$

Then

$$\mathcal{F}_H(S) + \{-b, 0, b\} = \mathcal{F}_H(S'). \quad (3.7)$$

Here $X + Y = \{x + y : x \in X, y \in Y\}$. Consequently, if a dissociated prefix A_n with total S_n satisfies $\mathcal{D}(A_n) = \mathcal{F}_H(S_n)$, the next chosen term is $a(n+1) = S_n + c_n$, and $S_n > 2M + |c_n|$, then

$$\mathcal{D}(A_{n+1}) = \mathcal{F}_H(S_{n+1}), \quad S_{n+1} = 2S_n + c_n \quad (3.8)$$

(source 43's Corollary 3.3, "inductive signed-difference certificate").

Proof. By Lemma 2.3, (3.7) is (3.5) shifted by $-(S + b)$, so it suffices to check (3.4). We have $b \geq S - |c| > 2M > M$ and $b \leq S + |c| < 2S - 2M \leq 2S - M$. Also $2S - b = S - c \geq S - |c| > 2M \geq \max(H + H)$. The consequence follows because appending $a(n+1)$ changes the signed difference set by $\mathcal{D}(A_{n+1}) = \mathcal{D}(A_n) + \{-a(n+1), 0, a(n+1)\}$. $\square$

Second proof of (3.7) (source 43). First ignore the deleted points. The three translated intervals are

$$\begin{aligned} [-S, S] - b &= [-S', -c], \\ [-S, S] &= [-S, S], \\ [-S, S] + b &= [c, S']. \end{aligned}$$

Because $S > |c|$, their union is all of $[-S', S'] \cap \mathbb{Z}$.

We now track the holes. Consider the right boundary point $S' - h$ with $h \in H$. It is larger than S because $S + c - h > 0$ follows from (3.6). Hence only the right translate can reach it. Its preimage there is $S - h$, which is deleted from $\mathcal{F}_H(S)$. Therefore $S' - h$ is absent. By symmetry, $-S' + h$ is absent as well.

It remains to show that no other point is lost. If $x \in [-S, S]$ is not a central hole, the untranslated copy contains it. If $x = S - h$ is a central hole, then in the right translate it has preimage

$$x - b = (S - h) - (S + c) = -c - h.$$

Now

$$|c + h| \leq |c| + M < S - M.$$

Every deleted point of $\mathcal{F}_H(S)$ has absolute value at least $S - M$, so $-c - h$ is present; hence x is restored by the right translate. The negative central holes are restored symmetrically by the left translate.

Finally suppose $S < x \leq S'$. Only the right translate can reach x , and its preimage $y = x - b$ lies in $(-c, S]$. A negative hole has the form $-S + h$ and satisfies

$$-S + h \leq -S + M < -|c| \leq -c,$$

so a missing preimage in $(-c, S]$ can only be a positive hole $S - h$. This gives exactly $x = S' - h$, already listed. The region $[-S', -S]$ is symmetric. Therefore the only missing points are the prescribed boundary defects. $\square$

[Merge note, batch 7301.] Source 43 writes the new term as $a = S - d$ with hypothesis $S > 2M + |d|$; with $c = -d$ this is (3.6). Its hypothesis implies source 50's conditions (3.4), but not conversely: source 50's theorem also covers terms b outside the window $|b - S| < S - 2M$. The intake tested source 50's theorem in 11,984 random fringe configurations and source 43's in 9,700, with no failure; source 43's hypothesis implied source 50's in every case. In all 1,794 tested cases with $M < b \leq 2S - M$ but $2S - b \in H + H$ the fringe broke, so the collision condition cannot be dropped.

Why this lemma is reusable (source 43)

The theorem isolates a finite-state mechanism. Once a greedy difference set becomes a full interval with a fixed collection of boundary defects, the next admissible term often has the form $S_n + c_n$, where c_n depends only on a residue state. If c_n is periodic, then $S_{n+1} = 2S_n + c_n$ is exactly solvable and the generated sequence is dyadic with a periodic correction. Sections 4 and 5 are two concrete realizations.

3.3 Greedy choice from a fixed fringe

Fix integers $m \geq 2$ and $r \in \{0, 1, \dots, m-1\}$. Suppose every selected term is congruent to r modulo m .

Proposition 3.5 (finite residue rule). *Assume (3.1) with total $S > M$. For each $h \in H$, the number $S - h$ is admissible, and every other positive integer at most S is forbidden. Therefore the next greedy term congruent to r modulo m is*

$$b = \min(\{S - h : h \in H, S - h \equiv r \pmod{m}\} \cup \{S + q\}), \quad (3.9)$$

where $q \in \{1, \dots, m\}$ is the least positive integer satisfying $S + q \equiv r \pmod{m}$.

Proof. Equation (3.2) lists all admissible positive integers in $[1, S]$. Every integer above S is outside $\mathcal{D}(A)$ and hence admissible by Lemma 2.1; $S + q$ is the first such integer in the required residue class. $\square$

If several h have the required residue, the largest h wins because it gives the smallest $S - h$. The correction

$$c(S) := b - S \quad (3.10)$$

is therefore selected from the finite set $\{-h : h \in H\} \cup \{1, \dots, m\}$ and depends only on $S \bmod m$.

3.4 Why a periodic correction gives a linear recurrence

Let $S_n = \sum_{j \leq n} a(j)$. Since every term is r modulo m ,

$$S_n \equiv nr \pmod{m}. \quad (3.11)$$

The residue cycle has period

$$p = \frac{m}{\gcd(m, r)}. \quad (3.12)$$

Under a permanent fixed fringe, Proposition 3.5 therefore gives a p -periodic sequence c_n such that

$$a(n+1) = S_n + c_n. \quad (3.13)$$

Theorem 3.6 (periodic correction principle). *Suppose (3.13) holds and c_n is periodic with period p from some point onward. Then*

$$d_n := a(n+1) - 2a(n) = c_n - c_{n-1} \quad (3.14)$$

is periodic with period p and has zero sum over each period. Consequently,

$$a(n+p) = a(n+p-1) + a(n+p-2) + \dots + a(n+1) + 2a(n) \quad (3.15)$$

from the corresponding eventual range onward. Its characteristic polynomial is

$$(x-2)(1+x+\dots+x^{p-1}) = x^p - x^{p-1} - \dots - x - 2. \quad (3.16)$$

Proof. Using $S_n = S_{n-1} + a(n)$ and $a(n) = S_{n-1} + c_{n-1}$ gives

$$a(n+1) - 2a(n) = S_n + c_n - 2a(n) = c_n - c_{n-1}.$$

The sum of p consecutive first differences of the periodic sequence c_n is zero. Hence

$$0 = \sum_{j=0}^{p-1} (a(n+j+1) - 2a(n+j)),$$

which simplifies to (3.15). Expanding the product gives (3.16). $\square$

A further useful consequence is an exact exponential-plus-periodic form. Since the roots other than 2 in (3.16) are nontrivial p th roots of unity, every solution is

$$a(n) = C2^n + p_n, \tag{3.17}$$

where p_n is p -periodic. In the two OEIS cases $p = 3$.

Source 43 proves the same principle in partial-sum coordinates, where the greedy step (3.13) reads $S_{n+1} = 2S_n + c_n$. Its statement adds the uniqueness of the periodic part and the denominator of the generating function.

Proposition 3.7 (periodic forcing principle; sources 43 and 50). *Suppose that, from some index onward,*

$$S_{n+1} = 2S_n + c_n, \quad c_{n+p} = c_n \tag{3.18}$$

for a fixed period p . Then there is a unique p -periodic sequence q_n satisfying

$$q_{n+1} = 2q_n + c_n,$$

and a constant κ such that

$$S_n = \kappa 2^n + q_n. \tag{3.19}$$

Consequently, $a(n) = S_n - S_{n-1}$ is a dyadic exponential plus a period- p sequence of mean zero, in agreement with (3.17) (with $C = \kappa/2$). Its ordinary generating function is rational, with denominator dividing

$$(1 - 2x)(1 + x + \cdots + x^{p-1}). \tag{3.20}$$

Proof (source 43). Iterating the affine recurrence for q over one period sends q_n to $2^p q_n + B_n$, where B_n is determined by the p forcing values. A periodic solution must satisfy $(1 - 2^p)q_n = B_n$, so it exists and is unique. Subtracting this periodic solution from (3.18) leaves $S_{n+1} - q_{n+1} = 2(S_n - q_n)$, proving (3.19).

Now $a(n) = \kappa 2^{n-1} + q_n - q_{n-1}$. The periodic difference has mean zero over one period, so its generating-function denominator divides $(1 - x^p)/(1 - x) = 1 + x + \cdots + x^{p-1}$. The dyadic part contributes $1 - 2x$. $\square$

For $p = 3$ the denominator is $(1 - 2x)(1 + x + x^2) = 1 - x - x^2 - 2x^3$; equivalently the characteristic polynomial is $(x - 2)(x^2 + x + 1) = x^3 - x^2 - x - 2$, which is exactly the recurrence polynomial appearing in both OEIS entries.

4 OEIS A139217: the residue-one case modulo 3

4.1 The seed fringe

The first three greedy terms are 1, 4, 7. Indeed, after 1 the values 1 and -1 are forbidden, so the next positive integer congruent to 1 modulo 3 is 4; after $\{1, 4\}$ the positive signed differences are 1, 3, 4, 5, so the next permitted value is 7.

Set $A_3 = \{1, 4, 7\}$ and $S_3 = 12$. From

$$(1 + x + x^2)(1 + x^4 + x^8)(1 + x^7 + x^{14})$$

one finds

$$\mathcal{E}(A_3) = ([0, 24] \cap \mathbb{Z}) \setminus \{3, 21\}. \quad (4.1)$$

For completeness, the attainable values in $[0, 12]$ are

$$0, 1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12;$$

reflection about 12 supplies the upper half. Thus the fringe is

$$H_{217} = \{3\}. \quad (4.2)$$

Signed certificate (source 43)

In signed coordinates (4.1) reads

$$\mathcal{D}(1, 4, 7) = [-12, 12] \cap \mathbb{Z} \setminus \{\pm 9\} = \mathcal{F}_{\{3\}}(12). \quad (4.3)$$

The positive integers 1, $\dots$, 12 other than 9 have the signed representations

$$\begin{aligned} 1 &= 1, & 2 &= 7 - 4 - 1, & 3 &= 4 - 1, \\ 4 &= 4, & 5 &= 4 + 1, & 6 &= 7 - 1, \\ 7 &= 7, & 8 &= 7 + 1, & 10 &= 7 + 4 - 1, \\ 11 &= 7 + 4, & 12 &= 7 + 4 + 1. \end{aligned}$$

The value 9 is impossible: with coefficient 0 on 7 the absolute value is at most 5; with coefficient 1 on 7, one would need a signed combination of 1 and 4 equal to 2, which does not occur; coefficient -1 cannot produce a positive value as large as 9.

4.2 The residue table

Every term is 1 modulo 3, so $S_n \equiv n \pmod{3}$. The only positive gap at most S_n is $S_n - 3$, which has the same residue as S_n . Proposition 3.5 gives the following exact table.

$n \bmod 3$	$S_n \bmod 3$	next admissible $a(n+1)$	$c_n = a(n+1) - S_n$	$d_n = a(n+1) - 2a(n)$
0	0	$S_n + 1$	1	-1
1	1	$S_n - 3$	-3	-4
2	2	$S_n + 2$	2	5

The d_n column is $c_n - c_{n-1}$, with residues read cyclically. Source 43 derives the same table: when $n \equiv 1 \pmod{3}$ the unique positive hole $S_n - 3$ is itself congruent to 1 and is selected; in the other two cases the hole has the wrong residue, every admissible positive integer at most S_n is represented, and the least admissible integer above S_n is $S_n + 1$ or $S_n + 2$.

4.3 Persistence of the fringe

Here $M = 3$, $m = 3$, and $S_3 = 12 > 2M + m = 9$. Every candidate in the table has correction $c \in [-3, 3]$. Corollary 3.3 applies immediately. Therefore the fringe (4.2) persists for every $n \geq 3$. (In source 43's form: $12 > 2 \cdot 3 + 3$, so Corollary 3.4 applies from the base case onward for all $c_n \in \{1, -3, 2\}$.)

Theorem 4.1 (complete solution of A139217; sources 50 and 43). *Let $a(n)$ be OEIS A139217 and $S_n = \sum_{j \leq n} a(j)$. Then:*

(a) For every $n \geq 3$,

$$\mathcal{E}(A_n) = [0, 2S_n] \cap \mathbb{Z} \setminus \{3, 2S_n - 3\},$$

and

$$\mathcal{D}_n = [-S_n, S_n] \cap \mathbb{Z} \setminus \{\pm(S_n - 3)\}.$$

(b) For every $n \geq 3$,

$$a(n+1) = 2a(n) + \begin{cases} -1, & n \equiv 0 \pmod{3}, \\ -4, & n \equiv 1 \pmod{3}, \\ 5, & n \equiv 2 \pmod{3}. \end{cases} \quad (4.4)$$

(c) For every $n \geq 6$,

$$a(n) = a(n-1) + a(n-2) + 2a(n-3). \quad (4.5)$$

(d) For every $n \geq 3$,

$$a(n) = 3 \cdot 2^{n-2} + \begin{cases} 1, & n \equiv 0, 1 \pmod{3}, \\ -2, & n \equiv 2 \pmod{3}. \end{cases} \quad (4.6)$$

Equivalently,

$$a(n) = \frac{3}{4} 2^n - 2 \cos\left(\frac{2\pi(n+1)}{3}\right). \quad (4.7)$$

(e) The ordinary generating function $A_{217}(x) = \sum_{n \geq 1} a(n)x^n$ is

$$A_{217}(x) = \frac{x + 3x^2 + 2x^3 - 6x^5}{1 - x - x^2 - 2x^3}. \quad (4.8)$$

(f) For $n \geq 3$,

$$S_n = 3 \cdot 2^{n-1} + \begin{cases} 0, & n \equiv 0 \pmod{3}, \\ 1, & n \equiv 1 \pmod{3}, \\ -1, & n \equiv 2 \pmod{3}. \end{cases} \quad (4.9)$$

Proof. Part (a) follows by induction from (4.1) and Theorem 3.2, as justified above. The residue table then gives the three-periodic corrections $c_n = (1, -3, 2)$ indexed by $n \equiv (0, 1, 2)$. Equation (3.14) gives (b).

The values d_n in (b) sum to zero over a period. Summing three consecutive relations $a(k+1) - 2a(k) = d_k$ gives (c), beginning with $n = 6$ because the periodic law already holds for d_3, d_4, d_5 .

For (d), the claimed right side agrees with $a(3) = 7$, and its periodic correction $p_n \in \{1, 1, -2\}$ satisfies $p_{n+1} - 2p_n = (-1, -4, 5)$ in the corresponding residue classes. Induction using (b) proves the formula. The cosine identity is the elementary period-three interpolation of $(1, 1, -2)$.

Multiplying $\sum_{n \geq 1} a(n)x^n$ by $1 - x - x^2 - 2x^3$ and using (c) leaves only the coefficients through degree five, which are $1, 3, 2, 0, -6$; this gives (e). Finally, summing (d), or checking the three residue classes in $S_n = S_{n-1} + a(n)$, proves (f). $\square$

In source 43's notation, part (a) with the greedy step reads: for every $n \geq 3$, $u_{n+1} = U_n + c_n$ with $c_n = (1, -3, 2)$, hence

$$U_{n+1} = 2U_n + c_n, \quad (4.10)$$

and part (f) is its Theorem 4.2, $U_n = 3 \cdot 2^{n-1} + q_n$ with $q_n = (0, 1, -1)$ for $n \equiv (0, 1, 2) \pmod{3}$.

Second proof of (a), (d), (f) and (c) (source 43). Equation (4.3) is the base case. Assuming (a) at stage n , the residue analysis proves that the greedy algorithm selects $u_{n+1} = U_n + c_n$; the threshold condition holds, so Corollary 3.4 propagates the same defect set to stage $n + 1$. This simultaneously proves the greedy choice and the invariant by induction.

The periodic forcing in (4.10) can be solved without guessing. A direct check of the three residue classes shows $q_{n+1} = 2q_n + c_n$. Since $U_3 = 12 = 3 \cdot 2^2 + q_3$, induction gives (f). Taking $u_n = U_n - U_{n-1}$ gives the piecewise formula (d), including $n = 3$ by direct substitution; the trigonometric form is the real Fourier representation of the period-three correction.

For (c), let τ be the forward shift. The correction term p_n is annihilated by $\tau^2 + \tau + 1$, while 2^n is annihilated by $\tau - 2$, so u_n is annihilated by $(\tau - 2)(\tau^2 + \tau + 1) = \tau^3 - \tau^2 - \tau - 2$. The closed form is valid from $n = 3$, so applying its order-three annihilator requires $n - 3 \geq 3$, namely $n \geq 6$. At $n = 5$ the two sides of (4.5) are 22 and 28. $\square$

Source 43 states (c) as its Corollary 4.3 and adds: “This range is sharp for the displayed initial sequence: the identity fails at $n = 5$.”

[Merge note, batch 73O1.] The recurrence also holds at $n = 4$ ($13 = 7 + 4 + 2 \cdot 1$), fails at $n = 5$, and holds for every $n \geq 6$; 6 is the first index from which it holds permanently. “Sharp” is true in that sense only.

4.4 Correction to the current OEIS range

As of 1 October 2026, A139217 states that the recurrence holds for $n > 4$. At $n = 5$ the proposed identity would read

$$22 \stackrel{?}{=} 13 + 7 + 2 \cdot 4 = 28,$$

so that range is false. Theorem 4.1 shows that the correct range is $n \geq 6$. The conjectured recurrence itself is correct after that single indexing correction. The eventual doubling correction recorded in the OEIS entry is now exact: it is (4.4) (sources 50 and 43).

5 OEIS A139218: the residue-two case modulo 3

5.1 The seed fringe

The first three terms are 2, 5, 8. For $A_3 = \{2, 5, 8\}$, $S_3 = 15$, and

$$\mathcal{E}(A_3) = ([0, 30] \cap \mathbb{Z}) \setminus \{1, 3, 6, 11, 19, 24, 27, 29\}. \quad (5.1)$$

Indeed, the represented values in $[0, 15]$ are

$$0, 2, 4, 5, 7, 8, 9, 10, 12, 13, 14, 15,$$

and reflection about 15 gives the upper half. Hence

$$H_{218} = \{1, 3, 6, 11\}. \quad (5.2)$$

Signed certificate (source 43)

The positive represented signed values for $\{2, 5, 8\}$ are

$$\begin{aligned} 1 &= 8 - 5 - 2, & 2 &= 2, & 3 &= 5 - 2, \\ 5 &= 5, & 6 &= 8 - 2, & 7 &= 5 + 2, \\ 8 &= 8, & 10 &= 8 + 2, & 11 &= 8 + 5 - 2, \\ 13 &= 8 + 5, & 15 &= 8 + 5 + 2. \end{aligned}$$

The four remaining positive integers are impossible. Indeed, $\mathcal{D}(2, 5) = \{0, \pm 2, \pm 3, \pm 5, \pm 7\}$; after fixing the coefficient of 8 to be 0, 1, or -1 , none of 4, 9, 12, 14 can be completed by an element of this set. Hence

$$\mathcal{D}(2, 5, 8) = [-15, 15] \cap \mathbb{Z} \setminus \{\pm 4, \pm 9, \pm 12, \pm 14\}, \quad (5.3)$$

and $\{4, 9, 12, 14\} = \{15 - h : h \in H_{218}\}$.

5.2 The residue table

Now every term is 2 modulo 3, so $S_n \equiv 2n \pmod{3}$. A gap $S_n - h$ is congruent to 2 when

$$h \equiv S_n - 2 \equiv 2(n - 1) \pmod{3}.$$

If several fringe elements match, the largest one yields the smallest greedy candidate. This gives:

$n \bmod 3$	$S_n \bmod 3$	matching h	chosen h	c_n	d_n
0	0	1	1	-1	10
1	2	3, 6	6	-6	-5
2	1	11	11	-11	-5

Thus the next term is always a fringe gap; no above- S_n fallback is needed. Source 43's table ("least admissible hole" $v_{n+1} = V_n - 1, V_n - 6, V_n - 11$) is the same.

5.3 Persistence of the fringe

Here $M = 11$. At the initial stage $S_3 = 15$ and the table gives $b = 14$. The conditions of Theorem 3.2 hold directly:

$$11 < 14 \leq 19, \quad 2S_3 - b = 16 \notin H_{218} + H_{218},$$

where

$$H_{218} + H_{218} = \{2, 4, 6, 7, 9, 12, 14, 17, 22\}. \quad (5.4)$$

After adjoining 14, the new total is $S_4 = 29 > 2M + 3 = 25$. Corollary 3.3 then applies forever.

The step $n = 3 \rightarrow 4$ by hand (source 43)

Source 43's threshold form does not cover the first step ($S_3 = 15 < 2 \cdot 11 + 1$), so it computes $\mathcal{D}(2, 5, 8, 14)$ directly. Appending 14 replaces the signed set by $\mathcal{D}_3 + \{-14, 0, 14\}$. On the positive side, the translate by 14 restores the four central holes because their preimages are $-10, -5, -2, 0 \in \mathcal{D}_3$. Every integer from 16 through 29 has only the preimage $x - 14 \in [2, 15]$; it is missing precisely when that preimage is one of 4, 9, 12, 14. Symmetry now gives

$$\mathcal{D}(2, 5, 8, 14) = [-29, 29] \cap \mathbb{Z} \setminus \{\pm 18, \pm 23, \pm 26, \pm 28\} = \mathcal{F}_{H_{218}}(29). \quad (5.5)$$

Thus both finite base identities have explicit hand-checkable certificates; the accompanying program `code/43-boundary-verify.py` independently verifies them by exact set equality. For the induction step from $n = 4$, the threshold (3.6) holds: $M = 11$, $S_4 = 29$, and for the first step $|c_4| = 6$, so $29 > 22 + 6$; thereafter the partial sums increase, and the largest possible $|c_n|$ is 11.

Theorem 5.1 (complete solution of A139218; sources 50 and 43). *Let $a(n)$ be OEIS A139218 and $S_n = \sum_{j \leq n} a(j)$. Then:*

(a) For every $n \geq 3$,

$$\mathcal{E}(A_n) = [0, 2S_n] \cap \mathbb{Z} \setminus (H_{218} \cup (2S_n - H_{218})),$$

where $H_{218} = \{1, 3, 6, 11\}$. Equivalently,

$$\mathcal{D}_n = [-S_n, S_n] \cap \mathbb{Z} \setminus \{\pm(S_n - h) : h \in H_{218}\}.$$

(b) For every $n \geq 4$,

$$a(n+1) = 2a(n) + \begin{cases} 10, & n \equiv 0 \pmod{3}, \\ -5, & n \equiv 1 \pmod{3}, \\ -5, & n \equiv 2 \pmod{3}. \end{cases} \quad (5.6)$$

(c) For every $n \geq 7$,

$$a(n) = a(n-1) + a(n-2) + 2a(n-3). \quad (5.7)$$

(d) For every $n \geq 4$,

$$a(n) = \frac{39}{56} 2^n + \begin{cases} -25/7, & n \equiv 0 \pmod{3}, \\ 20/7, & n \equiv 1 \pmod{3}, \\ 5/7, & n \equiv 2 \pmod{3}. \end{cases} \quad (5.8)$$

Equivalently, in an explicitly integral piecewise form,

$$a(n) = \begin{cases} (39 \cdot 2^n - 200)/56, & n \equiv 0 \pmod{3}, \\ (39 \cdot 2^n + 160)/56, & n \equiv 1 \pmod{3}, \\ (39 \cdot 2^n + 40)/56, & n \equiv 2 \pmod{3}, \end{cases} \quad (5.9)$$

that is, $a(n) = (39 \cdot 2^{n-3} - 25)/7$, $(39 \cdot 2^{n-3} + 20)/7$, $(39 \cdot 2^{n-3} + 5)/7$ respectively (source 43's form).

(e) The ordinary generating function $A_{218}(x) = \sum_{n \geq 1} a(n)x^n$ is

$$A_{218}(x) = \frac{2x + 3x^2 + x^3 - 3x^4 - 9x^5 - 12x^6}{1 - x - x^2 - 2x^3}. \quad (5.10)$$

(f) For every $n \geq 3$,

$$S_n = \frac{39}{28} 2^n + \begin{cases} 27/7, & n \equiv 0 \pmod{3}, \\ 47/7, & n \equiv 1 \pmod{3}, \\ 52/7, & n \equiv 2 \pmod{3}. \end{cases} \quad (5.11)$$

Proof. The direct base check and the propagation argument prove (a). The residue table gives $c_n = (-1, -6, -11)$ for $n \equiv (0, 1, 2)$, respectively. Hence $d_n = c_n - c_{n-1} = (10, -5, -5)$ once both adjacent corrections are in the stable regime, namely for $n \geq 4$. This proves (b).

The three values of d_n sum to zero, so Theorem 3.6 gives (c), starting at $n = 7$. For (d), the periodic correction $(-25/7, 20/7, 5/7)$ has successive doubled differences $(10, -5, -5)$, and the formula agrees with $a(4) = 14$; induction using (b) completes the proof.

Multiplication by $1 - x - x^2 - 2x^3$ leaves the six initial residual coefficients $2, 3, 1, -3, -9, -12$, giving (e). Summation of (d), or induction in the three residue classes, gives (f). $\square$

In source 43's notation, part (a) with the greedy step reads: for every $n \geq 3$, $v_{n+1} = V_n + c_n$ with $c_n = (-1, -6, -11)$, hence

$$V_{n+1} = 2V_n + c_n, \quad (5.12)$$

and part (f) is its Theorem 5.2, $V_n = \frac{39}{28}2^n + s_n$, with the period-three sequence

$$s_n = \begin{cases} 27/7, & n \equiv 0 \pmod{3}, \\ 47/7, & n \equiv 1 \pmod{3}, \\ 52/7, & n \equiv 2 \pmod{3}, \end{cases} \quad (5.13)$$

and $v_n = \frac{39}{56}2^n + r_n$, $r_n = (-25/7, 20/7, 5/7)$.

Second proof of (a), (d), (f) and (c) (source 43). The cases $n = 3$ and $n = 4$ are the direct computations (5.3) and (5.5); the transition $n = 3$ to $n = 4$ also agrees with the greedy step. For $n \geq 4$, the residue table identifies the greedy term, and Corollary 3.4 propagates the fixed defect set. Induction completes the proof of (a).

One checks in the three residue classes that $s_{n+1} = 2s_n + c_n$. Since $V_4 = 29 = \frac{39}{28}2^4 + \frac{47}{7}$, the first-order recurrence (5.12) gives (f) by induction. Taking the difference $V_n - V_{n-1}$ and simplifying gives (d); integrality also follows automatically because the closed form equals the integer sequence generated by the greedy construction. For (c), the dyadic term and the period-three correction are annihilated by $(\tau - 2)(\tau^2 + \tau + 1)$; the closed form begins at $n = 4$, so the recurrence begins at $n = 7$. $\square$

[Merge note, batch 7301.] Source 43 states (f) for $n \geq 4$ only, starting the induction at $V_4 = 29$. The formula also holds at $n = 3$: $\frac{39}{28} \cdot 8 + \frac{27}{7} = 15 = V_3$ (source 50 states $n \geq 3$). Source 43's delivered OEIS draft (43-boundary-OEIS_update_draft.txt, shipped unchanged) keeps "For $n \geq 4$ ".

The current OEIS range $n > 6$ for (5.7) is therefore correct. The exact doubling correction (5.6) proves the eventual periodicity observed in the OEIS entry.

6 Relations between the two sequences (source 43)

6.1 Cross-sequence identities

The dominant constants satisfy

$$\frac{39/56}{3/4} = \frac{13}{14}.$$

Subtracting the two closed forms (4.6) and (5.8) gives a stronger exact statement.

Corollary 6.1 (cross-sequence identity). *For every $n \geq 4$,*

$$14v_n - 13u_n = \begin{cases} -63, & n \equiv 0 \pmod{3}, \\ 27, & n \equiv 1 \pmod{3}, \\ 36, & n \equiv 2 \pmod{3}. \end{cases} \quad (6.1)$$

In particular,

$$\frac{v_n}{u_n} = \frac{13}{14} + O(2^{-n}).$$

Proof. For $n \geq 4$ both closed forms hold, and $14 \cdot \frac{39}{56} = 13 \cdot \frac{3}{4}$, so the dyadic terms cancel. The periodic remainders are $14 \cdot (-\frac{25}{7}) - 13 \cdot 1 = -63$, $14 \cdot \frac{20}{7} - 13 \cdot 1 = 27$ and $14 \cdot \frac{5}{7} - 13 \cdot (-2) = 36$. $\square$

The range is sharp: at $n = 3$, $14 \cdot 8 - 13 \cdot 7 = 21 \neq -63$. A corresponding identity for partial sums is

$$14V_n - 13U_n = \begin{cases} 54, & n \equiv 0 \pmod{3}, \\ 81, & n \equiv 1 \pmod{3}, \\ 117, & n \equiv 2 \pmod{3}, \end{cases} \quad (n \geq 4), \quad (6.2)$$

by the same subtraction applied to (4.9) and (5.11).

[**Merge note, batch 73O1.**] Since both partial-sum formulas hold from $n = 3$, (6.2) also holds at $n = 3$: $14 \cdot 15 - 13 \cdot 12 = 54$.

6.2 Near-saturation of the signed-difference interval

The structural theorems immediately give the cardinalities

$$|\mathcal{D}_n| = 2U_n - 1 \quad (\text{A139217}), \quad (6.3)$$

$$|\mathcal{D}_n| = 2V_n - 7 \quad (\text{A139218}), \quad (6.4)$$

for every $n \geq 3$. The ambient intervals have $2U_n + 1$ and $2V_n + 1$ points, respectively. Hence the support of the signed differences misses only 2 points in the first sequence and 8 in the second, independently of n . This fixed-defect saturation is much stronger information than the term recurrence.

7 Exact asymptotics and transseries

7.1 Unified exponential-plus-periodic form

Both solutions have the exact form

$$a(n) = C2^n + p_j, \quad j \equiv n \pmod{3}, \quad (7.1)$$

with the following constants.

Sequence	valid range	C	(p_0, p_1, p_2)	leading asymptotic
A139217	$n \geq 3$	$3/4$	$(1, 1, -2)$	$a(n) \sim (3/4)2^n$
A139218	$n \geq 4$	$39/56$	$(-25/7, 20/7, 5/7)$	$a(n) \sim (39/56)2^n$

The periodic term is bounded, so there are no algebraic corrections in $1/n$. The complete correction scale is exponentially small relative to the main term.

Corollary 7.1 (forward asymptotics; source 43). *As $n \rightarrow \infty$,*

$$\begin{aligned} u_n &= \frac{3}{4} 2^n + O(1), & U_n &= \frac{3}{2} 2^n + O(1), \\ v_n &= \frac{39}{56} 2^n + O(1), & V_n &= \frac{39}{28} 2^n + O(1). \end{aligned}$$

The errors are exactly periodic of period 3. Moreover, $u_{n+1}/u_n = 2 + O(2^{-n})$ and $v_{n+1}/v_n = 2 + O(2^{-n})$.

The closed forms are already complete asymptotic expansions: there are no uncomputed algebraic or logarithmic correction terms. They can also be viewed as exact exponential polynomials (source 43). With $\omega = e^{2\pi i/3}$,

$$u_n = \frac{3}{4} 2^n - \omega^{n+1} - \omega^{-(n+1)},$$

and the period-three term of v_n is

$$r_n = -\frac{25}{7} \cos \frac{2\pi n}{3} + \frac{5\sqrt{3}}{7} \sin \frac{2\pi n}{3}.$$

Thus the complete spectral support consists of the three characteristic roots $2, \omega, \bar{\omega}$.

7.2 Logarithmic transseries

For a fixed residue branch j and all sufficiently large $n \equiv j \pmod{3}$,

$$\begin{aligned} \log a(n) &= n \log 2 + \log C + \log \left(1 + \frac{p_j}{C2^n} \right) \\ &= n \log 2 + \log C + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \left(\frac{p_j}{C} \right)^k 2^{-kn}. \end{aligned} \quad (7.2)$$

This is not merely asymptotic: it is a convergent expansion once $|p_j| < C2^n$, which already holds throughout the closed-form ranges above.

Similarly, if d_j is the period-three value of $a(n+1) - 2a(n)$ on the branch $n \equiv j$, then

$$\begin{aligned} \frac{a(n+1)}{a(n)} &= 2 + \frac{d_j}{C2^n + p_j} \\ &= 2 + \frac{d_j}{C} 2^{-n} \sum_{k \geq 0} \left(-\frac{p_j}{C} \right)^k 2^{-kn}. \end{aligned} \quad (7.3)$$

Thus convergence to the growth ratio 2 has an explicit oscillatory period-three expansion to all orders in 2^{-n} .

7.3 Inverse transseries on residue branches

Because of the periodic correction, the inverse is naturally three-branched. For a fixed residue j , solve $x = C2^n + p_j$ continuously for n :

$$\nu_j(x) := \log_2 \left(\frac{x - p_j}{C} \right). \quad (7.4)$$

Whenever n lies in the closed-form range and $n \equiv j \pmod{3}$,

$$\nu_j(a(n)) = n \quad (7.5)$$

exactly. Expanding at infinity gives the logarithmic inverse transseries

$$\nu_j(x) = \log_2 \left(\frac{x}{C} \right) - \frac{1}{\log 2} \sum_{k \geq 1} \frac{p_j^k}{k x^k}. \quad (7.6)$$

For A139217 the leading inverse is $\log_2(4x/3)$; for A139218 it is $\log_2(56x/39)$. Formula (7.6) supplies every inverse-power correction and keeps the residue dependence explicit. It was obtained independently by both sources; for $|x| > |p_j|$ it is not merely a formal Poincaré expansion but the convergent Taylor series of $\log(1 - p_j/x)$ (source 43).

Proposition 7.2 (truncation remainder; source 43). *Let $R_K(x)$ be the remainder after the first K terms of the sum in (7.6). For $|x| > |p_j|$,*

$$|R_K(x)| \leq \frac{|p_j|^{K+1}}{(K+1) \log 2 |x|^{K+1}} \frac{1}{1 - |p_j/x|}. \quad (7.7)$$

Proof. $|R_K(x)| \leq \frac{1}{\log 2} \sum_{k \geq K+1} \frac{|p_j/x|^k}{k} \leq \frac{1}{(K+1) \log 2} \sum_{k \geq K+1} |p_j/x|^k$, a geometric series. $\square$

Explicit branches (source 43)

For A139217, $C = 3/4$ and the periodic shifts are $p_j = 1, 1, -2$ for index residues $j = 0, 1, 2$. Therefore

$$\nu_0(x) = \nu_1(x) = \log_2 \frac{4x}{3} - \frac{1}{\log 2} \left(\frac{1}{x} + \frac{1}{2x^2} + \frac{1}{3x^3} + \frac{1}{4x^4} + \cdots \right), \quad (7.8)$$

$$\nu_2(x) = \log_2 \frac{4x}{3} + \frac{1}{\log 2} \left(\frac{2}{x} - \frac{2}{x^2} + \frac{8}{3x^3} - \frac{4}{x^4} + \cdots \right). \quad (7.9)$$

When $x = u_n$ and the appropriate branch is chosen, $\nu_j(x) = n$ exactly.

For A139218, $C = 39/56$, with shifts $p_0 = -\frac{25}{7}$, $p_1 = \frac{20}{7}$, $p_2 = \frac{5}{7}$. The first two corrections are

j	$\nu_j(x) - \log_2(56x/39)$
0	$\frac{1}{\log 2} \left(\frac{25}{7x} - \frac{625}{98x^2} + O(x^{-3}) \right)$
1	$-\frac{1}{\log 2} \left(\frac{20}{7x} + \frac{200}{49x^2} + O(x^{-3}) \right)$
2	$-\frac{1}{\log 2} \left(\frac{5}{7x} + \frac{25}{98x^2} + O(x^{-3}) \right)$

Again, these are convergent for $|x| > |p_j|$.

7.4 Exact counting functions

The closed forms also invert discretely. Let

$$N_A(x) := \#\{n \geq 1 : a(n) \leq x\}.$$

Define, for real y ,

$$\Lambda(y) := \begin{cases} \max\{0, \lfloor \log_8 y \rfloor\}, & y > 0, \\ 0, & y \leq 0. \end{cases} \quad (7.10)$$

This counts positive integers m for which $8^m \leq y$. (Source 43 calls this function Φ .)

For A139217, the three branches $n = 3m, 3m + 1, 3m + 2$ give respectively

$$\frac{4(a(n) - 1)}{3} = 8^m, \quad \frac{2(a(n) - 1)}{3} = 8^m, \quad \frac{a(n) + 2}{3} = 8^m.$$

Therefore, for every real x (sources 50 and 43),

$$N_{217}(x) = \mathbf{1}_{x \geq 1} + \mathbf{1}_{x \geq 4} + \Lambda\left(\frac{4(x-1)}{3}\right) + \Lambda\left(\frac{2(x-1)}{3}\right) + \Lambda\left(\frac{x+2}{3}\right). \quad (7.11)$$

For A139218, the residue-zero branch begins with $n = 6$ (that is, $m = 2$), while the other two closed-form branches begin with $m = 1$. Hence, for every real x ,

$$N_{218}(x) = \mathbf{1}_{x \geq 2} + \mathbf{1}_{x \geq 5} + \mathbf{1}_{x \geq 8} + \Lambda\left(\frac{7x+25}{39}\right) + \Lambda\left(\frac{28x-80}{39}\right) + \Lambda\left(\frac{14x-10}{39}\right). \quad (7.12)$$

Equations (7.11)–(7.12) are exact, not merely asymptotic inversion formulas.

[Merge note, batch 7301.] Source 50 printed the residue-zero term of (7.12) as $\max\{0, \lfloor \log_8(\frac{56x+200}{39}) \rfloor - 1\}$. That expression equals $\Lambda(\frac{7x+25}{39})$ for $x > -25/7$, because $\frac{56x+200}{39} = 8 \cdot \frac{7x+25}{39}$, but it is undefined for $x \leq -25/7$, where the count is 0. The term printed here is source 43's, which is defined for every real x .

Proof (source 43). Besides the exceptional terms $u_1 = 1$ and $u_2 = 4$, (4.6) gives, for $k \geq 1$,

$$u_{3k} = \frac{3}{4} 8^k + 1, \quad u_{3k+1} = \frac{3}{2} 8^k + 1, \quad u_{3k+2} = 3 \cdot 8^k - 2.$$

For a branch $A8^k + B$ with $k \geq 1$, the number of terms not exceeding x is $\Lambda((x-B)/A)$. Summing the three branches and the two exceptions gives (7.11). Besides $v_1 = 2, v_2 = 5, v_3 = 8$, the three branches of A139218 may be written, for $k \geq 1$, as

$$v_{3k+3} = \frac{39}{7} 8^k - \frac{25}{7}, \quad v_{3k+1} = \frac{39}{28} 8^k + \frac{20}{7}, \quad v_{3k+2} = \frac{39}{14} 8^k + \frac{5}{7}.$$

Applying the same branch count proves (7.12). $\square$

Corollary 7.3 (counting asymptotics; source 43).

$$N_{217}(x) = \log_2 x + O(1), \quad N_{218}(x) = \log_2 x + O(1) \quad (x \rightarrow \infty).$$

The $O(1)$ term cannot be replaced by a convergent constant: it contains bounded logarithmically periodic staircase oscillations produced by the three floor functions.

7.5 Two notions of inversion (source 43)

There are two complementary notions of inversion:

- the exact discrete counting functions (7.11) and (7.12), which retain floors and bounded log-periodic oscillation;
- the smooth residue-resolved inverses (7.6), which admit all-orders convergent expansions.

Discarding the residue branch before inversion loses an $O(1)$ amount of information, while discarding the periodic additive term within a fixed branch loses only an explicitly expandable $O(x^{-1})$ correction to the index.

8 Generating functions and recurrence diagnostics

The generating functions (4.8) and (5.10) were found by both sources (source 43's Theorem 6.2, with variable z). The common denominator factors as

$$1 - x - x^2 - 2x^3 = (1 - 2x)(1 + x + x^2). \quad (8.1)$$

The pole at $x = 1/2$ produces the exponential term $C2^n$. The roots of $1 + x + x^2$ are the nontrivial cube roots of unity and produce the bounded period-three correction. This analytic factorization exactly mirrors the finite-fringe residue mechanism; the common denominator is not a numerical coincidence (Proposition 3.7).

For A139217, the failure of the recurrence at $n = 5$ is recorded by the term $-6x^5$ in the numerator. Beginning at degree six, multiplication by the denominator gives zero. For A139218, the transient lasts through degree six, and the recurrence starts at degree seven.

The first-order deviations make the onset especially transparent:

Sequence	initial deviations $a(n+1) - 2a(n)$	periodic tail
A139217	2, -1	-1, -4, 5, -1, -4, 5, ... from $n = 3$
A139218	1, -2, -2	-5, -5, 10, -5, -5, 10, ... from $n = 4$

This reproduces and proves the patterns recorded as conjectural comments in the OEIS entries.

9 Exact computational audit

9.1 Bitset model (source 50's verifier)

The program `code/50-finite-fringe-verify.py` represents the support of $P_A(x)$ in (2.4) by one nonnegative Python integer. Bit j is one exactly when $j \in \mathcal{E}(A)$. Appending b is the exact operation

$$B \mapsto B \text{ OR } (B \ll b) \text{ OR } (B \ll 2b). \quad (9.1)$$

No floating-point computation or probabilistic test is involved.

Given B and S , positive admissible values $x \leq S$ correspond exactly to missing bits at positions $S + x$. The program enumerates those missing bits, filters by the desired congruence, and compares them with the first permitted integer above S . Thus it reconstructs the greedy sequence independently of the closed forms.

9.2 Checks performed by source 50's verifier

Running

```
python code/50-finite-fringe-verify.py
```

performs the following exact checks.

- (i) It generates 22 terms of each sequence directly from the greedy rule.
- (ii) It matches the 14 terms currently printed by the OEIS.
- (iii) From $n = 3$ onward it verifies every missing exponent in $[0, 2S_n]$, not merely a sample, against the claimed fixed fringe.
- (iv) It checks the closed forms and recurrence on their stated ranges.
- (v) It expands both rational generating functions through degree 30 and checks every coefficient against the closed form.

A representative run ends with

```
A139217: verified 22 greedy terms ... exact fringe [3].
A139218: verified 22 greedy terms ... exact fringe [1, 3, 6, 11].
A139217: rational generating function checked through x^30.
A139218: rational generating function checked through x^30.
All exact checks passed.
```

It does not check the counting functions, the cross-identities or the failure at $n = 5$; source 43's verifier does.

9.3 Source 43's verifier

The program `code/43-boundary-verify.py` is a standard-library Python program, intentionally independent of symbolic algebra systems. All principal checks use exact integers and rational numbers. It performs the following tasks.

- (1) It generates the two sequences directly from the greedy definition, maintaining the exact signed difference set.

- (2) It verifies the published OEIS prefixes and extends each sequence to seventeen terms.
- (3) For every checked prefix, it proves by finite set equality that the signed difference set has exactly the boundary-defect form of Theorems 4.1(a) and 5.1(a).
- (4) It checks the closed forms and the correct recurrence ranges.
- (5) It explicitly confirms the failure of the A139217 recurrence at $n = 5$.
- (6) It tests the boundary-propagation identity for all forcing values used in the proofs and several partial sums.
- (7) It checks the exact counting functions on every integer from 0 through the largest generated term.

The core update is only

```
def signed_difference_set(values):
    out = {0}
    for value in values:
        out = {x + eps * value
               for x in out for eps in (-1, 0, 1)}
    return out
```

The recorded output `data/43-boundary-verification_output.txt` begins:

```
All exact checks passed.
A139217 prefix: [1, 4, 7, 13, 22, 49, 97, 190, ...]
A139218 prefix: [2, 5, 8, 14, 23, 41, 92, 179, ...]
A139217 boundary offsets: {3}
A139218 boundary offsets: {1, 3, 6, 11}
```

9.4 Role and limits of computation

The finite checks are useful for detecting transcription and indexing errors. The proof of all n , however, is Theorem 3.2 plus the base fringes (4.1) and (5.1). Source 43 puts it the same way: the computation is useful for detecting the invariant and guarding against transcription errors, but a finite calculation cannot prove eventual behaviour. The infinite argument consists of three exact ingredients: a finite base difference-set identity; the greedy selection forced by residues and boundary holes; and the propagation theorem. Once these are proved, no extrapolation from data remains.

10 Proposed OEIS updates

Both packages contain a plain-text draft suitable for adaptation after independent review: `50-finite-fringe-oeis_update_draft.txt` and `43-boundary-OEIS_update_draft.txt`. Neither has been submitted. They agree on every formula and range. The mathematical content can be summarized as follows.

10.1 A139217

- Replace the conjectural period statement by the proved identity

$$a(n+1) - 2a(n) = (-1, -4, 5) \quad \text{periodically for } n \geq 3,$$

with the values ordered by $n \equiv 0, 1, 2 \pmod{3}$.

- Correct the recurrence range from “ $n > 4$ ” to “ $n \geq 6$ ”.
- Add the closed form (4.6), generating function (4.8), and exact signed-support statement from Theorem 4.1.

10.2 A139218

- Replace the conjectural period statement by the proved identity

$$a(n+1) - 2a(n) = (10, -5, -5) \text{ periodically for } n \geq 4,$$

with the values ordered by $n \equiv 0, 1, 2 \pmod{3}$.

- Confirm the recurrence for $n \geq 7$.
- Add the closed form (5.9), generating function (5.10), and exact signed-support statement from Theorem 5.1.

The support theorems are compact enough for an OEIS comment but stronger than what is needed to verify the numerical formula. A stable public manuscript link and checked attribution should be supplied before submission.

10.3 Source 43’s proposed wording

Source 43 offers the following text as a starting point for an OEIS submission, subject to independent review and attribution decisions.

*A139217. **Formula.*** For $n \geq 3$, $a(n) = 3 \cdot 2^{n-2} + 1$ if $n \equiv 0, 1 \pmod{3}$ and $a(n) = 3 \cdot 2^{n-2} - 2$ if $n \equiv 2 \pmod{3}$. Hence $a(n) = a(n-1) + a(n-2) + 2a(n-3)$ for $n \geq 6$. The previously displayed range $n > 4$ is false at $n = 5$. **Comment.** If $S_n = \sum_{k=1}^n a(k)$, then for every $n \geq 3$ the set of signed sums $\{\sum_{k=1}^n \varepsilon_k a(k) : \varepsilon_k \in \{-1, 0, 1\}\}$ is $[-S_n, S_n] \cap \mathbb{Z}$ with only $\pm(S_n - 3)$ removed. This invariant proves the greedy formula. **Generating function.** (4.8).

*A139218. **Formula.*** For $n \geq 4$, (5.8). Hence $a(n) = a(n-1) + a(n-2) + 2a(n-3)$ for $n \geq 7$. **Comment.** If $S_n = \sum_{k=1}^n a(k)$, then for every $n \geq 3$ the signed-sum set is $[-S_n, S_n] \cap \mathbb{Z}$ with precisely $\pm(S_n - 1)$, $\pm(S_n - 3)$, $\pm(S_n - 6)$, $\pm(S_n - 11)$ removed. **Generating function.** (5.10).

Source 43’s delivered draft file also gives the partial sums (4.9) and $S_n = \frac{39}{28}2^n + s_n$ (stated there for $n \geq 4$; true from $n = 3$), source 43’s integral form of (5.8), and a status and attribution note. Source 50’s draft adds the cosine form (4.7), the periodic deviations, the leading asymptotics and a proof idea.

11 Formalization roadmap for ProveIt

The proof is well suited to Lean because its only infinite step is an induction over finite integer intervals and finite set translations.

11.1 Suggested definitions

A direct formalization can use finite sets rather than polynomials:

$$\begin{aligned} \text{signedSums}(A) &:= \left\{ \sum_i \varepsilon_i a_i : \varepsilon_i \in \{-1, 0, 1\} \right\}, \\ \text{ternarySums}(A) &:= \left\{ \sum_i \delta_i a_i : \delta_i \in \{0, 1, 2\} \right\}. \end{aligned}$$

For a list or finset of naturals, the second set can be defined recursively by

$$E([]) = \{0\}, \quad E(a :: L) = E(L) \cup (a + E(L)) \cup (2a + E(L)).$$

This recursion avoids coefficient reasoning entirely.

11.2 Core lemmas

A compact development would prove, in order:

- (1) distinct subset sums $\Leftrightarrow$ no nontrivial signed zero relation;
- (2) admissibility of a new term $b \Leftrightarrow b \notin \mathcal{D}(A)$;
- (3) $\mathcal{E}(A) = S(A) + \mathcal{D}(A)$ and reflection symmetry;
- (4) the three-copy recursion (2.5);
- (5) Theorem 3.2 as a lemma about three translated integer intervals with finite deleted fringes;
- (6) the finite residue rule and periodic correction principle;
- (7) decidable base computations for $\{1, 4, 7\}$ and $\{2, 5, 8\}$;
- (8) the two inductions, closed forms, and recurrence identities.

The only delicate part is organizing the interval-overlap proof so that arithmetic automation sees the cases. One practical approach is to state a separate coverage lemma for I_0, I_1, I_2 and then discharge the possible hole alignments with finite-set membership in $H + H$. For the concrete fringes, all remaining arithmetic should be handled by `omega`, `norm_num`, and finite `decide` checks.

11.3 Proof-certificate boundary

The executable verifier is intentionally not presented as a formal proof. A Lean development should import neither the generated terms nor a large bitset certificate as an axiom. The finite seed supports may be proved by normalization or reflected interval membership, while the universal step must be the abstract fringe theorem. This keeps the trusted kernel small and the mathematics reusable for other OEIS sequences.

11.4 Source 43's roadmap

Source 43 proposes a compact Lean development with the following layers.

Core definitions. Represent a finite prefix as a list or finset of integers and define $\text{delta}(A) = \{\sum_{a \in A} \varepsilon_a a : \varepsilon_a \in \{-1, 0, 1\}\}$. For formal efficiency, one can instead define $\Sigma(A)$ first and set $\mathcal{D}(A) = \Sigma(A) - \Sigma(A)$; the latter makes the append criterion a direct statement about two translated finsets. Suggested declarations include

```
subsetSums : List Int -> Finset Int
signedDiffs : List Int -> Finset Int
dissociated : List Int -> Prop
boundaryModel : Finset Nat -> Int -> Finset Int
```

General theorem layer. `dissociated_cons_iff_not_mem_signedDiffs`, formalizing Lemma 2.1(iii); a translate identity for signed differences after appending one term; `boundaryModel_propagate`, formalizing Corollary 3.4; and interval-membership lemmas that separate the central and two outer regions. The propagation proof uses only linear integer arithmetic, finite-set extensionality, and elementary inequalities, so most local obligations should be amenable to `omega`, `linarith`, and simplification.

Sequence-specific layer. For A139217, certify the base equality for (1, 4, 7) and then prove the three residue cases. For A139218, certify the two finite bases (2, 5, 8) and (2, 5, 8, 14). These bases are small enough for a transparent theorem-by-theorem proof; a checked decision procedure may also be used if the repository’s trust policy permits it. After the structural induction, the formulas are routine recurrences over residue classes. The final recurrence follows from either direct substitution in the piecewise closed forms or an abstract lemma for dyadic-plus-periodic sequences.

Recommended theorem boundary. A strong formalization should make the main public theorems state the greedy semantics, not only the numerical recurrence. Proving that a recurrence-generated list matches the first several terms is insufficient; the key statement is that every next term is the least admissible integer in the required congruence class. The signed-difference invariant provides exactly that bridge.

[Merge note, batch 7301.] None of these declarations exists in ProveIt. The names above are the sources’ proposals; nothing in this report is formalized.

12 Further research questions

The fixed-fringe mechanism suggests a broader program rather than two isolated identities. Questions 1–10 are source 50’s; Section 12.1 lists source 43’s, with the overlaps marked.

Question 1 (classification by modulus and residue). *For which pairs (m, r) does the progression-restricted greedy dissociated sequence eventually acquire a fixed fringe? Can the fringe and the onset index be computed from (m, r) alone?*

The cases $r = 0$ reduce after scaling by m to an unrestricted problem, while nonzero residues can forbid exact doubling. The first unexplored step is a systematic census for $m = 4, 5, 6$ with proof-oriented support data.

Question 2 (a finite fringe automaton). *Can one define a finite-state transition system whose states are normalized hole sets near 0 and $2S$, and prove that every progression-restricted greedy orbit eventually enters a cycle?*

A naive state space is infinite because a hole can drift into the bulk. The central issue is to prove a uniform bound on relevant fringe width, or to find a counterexample with a growing fringe. The three-copy recursion gives a natural transfer operator on hole patterns.

Question 3 (necessary and sufficient propagation conditions). *Theorem 3.2 gives simple sufficient conditions. Can one classify exactly the values of b for which a given fringe H is preserved? What new fringe is produced when the collision $2S - b \in H + H$ occurs?*

The collision equation suggests a controlled “fringe surgery” calculus: aligned high and low holes can either persist, be filled by the third copy, or create a new finite cluster. An exact transition theorem could make the automaton question tractable.

Question 4 (eventual exponential-plus-periodic laws). *Is every fixed-fringe progression sequence eventually of the form $C2^n + p_n$ with periodic p_n ? More generally, if the normalized fringe is periodic rather than fixed, does one still obtain a rational generating function?*

For a genuinely periodic family of fringes, the correction c_n would remain periodic after augmenting the state by the fringe phase. This strongly suggests rationality, but the onset and state bound need proof.

Question 5 (higher representation multiplicity). *Replace dissociation by the condition that every subset sum has at most g representations, or replace digits $\{-1, 0, 1\}$ by $\{-q, \dots, q\}$. Is there an analogue of finite-fringe propagation with $2q + 1$ translated copies?*

The geometric part of the proof generalizes immediately to multiple copies, but overlap multiplicities and admissibility are subtler. This direction connects directly with the generalized dissociation questions considered in [5].

Question 6 (unions of residue classes). *What happens when candidates are allowed in a union $R \subset \mathbb{Z}/m\mathbb{Z}$ rather than one residue class? Can the greedy choice be described by a finite minimum over fringe gaps, and when does exact doubling reappear?*

This interpolates between the present constrained sequences and Dutta’s unrestricted greedy algorithm. It may reveal a sharp threshold between exact eventual doubling and periodic near-doubling.

Question 7 (robustness under seeds). *Fix (m, r) but begin with an arbitrary finite dissociated seed in the residue class. Which seeds converge to the same fringe cycle, and which produce different asymptotic constants C ?*

The constant C is determined by the transient together with the periodic correction. Basin-of-attraction questions may be finite-state once a fringe bound is available.

Question 8 (effective inverse theory). *For eventual laws $a(n) = C2^n + p_n$, can one give uniform exact formulas for the counting function, nearest-index map, and error bounds when only a recurrence or fringe automaton is known?*

The formulas in Section 7 show that the inverse problem naturally decomposes by residue branch. For longer periods, roots-of-unity filters and floor-log expressions should give a systematic answer.

Question 9 (formal library theorem). *Can Theorem 3.2 be formalized once as a reusable ProveIt lemma and applied automatically to a certificate consisting only of a seed fringe and a finite residue table?*

Such a theorem would turn many OEIS conjectures into small, auditable proof certificates rather than sequence-specific formal developments.

Question 10 (OEIS exploration). *Which existing OEIS entries record unexplained eventual recurrences arising from greedy distinct-subset-sum constructions? Can their recurrences be reverse-engineered into finite fringe sets?*

A useful search strategy is to inspect entries with comments of the form “ $a(n + 1) - 2a(n)$ appears periodic,” compute exact ternary-support holes, and test whether the holes stabilize after translation by the total sum.

12.1 Source 43’s questions

Source 43 numbers its questions consecutively; its numbers are kept. For a modulus $m \geq 2$ and residue r , the greedy dissociated sequence chooses the least positive integer congruent to r modulo m that preserves distinct subset sums.

Classification by modulus and residue (overlaps Question 1).

- 43.1 For which pairs (m, r) does the signed difference set eventually become a full interval with a fixed finite boundary-defect set?
- 43.2 When this happens, can the defect set be computed from a finite seed by a terminating algorithm with proof certificates?
- 43.3 Is eventual boundary-defect stabilization guaranteed whenever $\gcd(m, r) = 1$, or are there primitive counterexamples?

Preliminary exact experiments indicate that fixed defect sets occur for several further small pairs, but a systematic theorem requires more than numerical evidence.

A *finite-state classification theorem* (overlaps Questions 2–4). The present examples have a particularly rigid form: $\mathcal{D}_n = \mathcal{F}_H(S_n)$ and $S_{n+1} = 2S_n + c(S_n \bmod m)$.

- 43.4 Classify all finite sets H and residue-choice functions c for which the propagation equation (3.7) is compatible with a greedy rule.
- 43.5 Develop a finite automaton whose states encode the boundary defects and whose accepted cycles certify eventual recurrences.
- 43.6 Determine whether every ultimately periodic defect automaton yields a rational generating function, and characterize its minimal denominator.

Beyond a single residue class (overlaps Question 6).

- 43.7 Replace one residue class by a union of classes, or by an arbitrary periodic admissibility set. Does the next-step forcing become automatic, morphic, or eventually periodic?
- 43.8 Study admissible sets defined by valuations, digital constraints, or Beatty sequences. Which constraints preserve finite-state boundary dynamics?
- 43.9 Allow a weighted greedy objective, such as the least admissible term under a cost function rather than the usual order.

Robustness and seed dependence (overlaps Question 7).

- 43.10 If finitely many initial terms are changed while preserving dissociation and congruence, does the process converge to the same defect cycle, a translated cycle, or one of finitely many attractors?
- 43.11 Quantify a basin of attraction for each stable defect set.
- 43.12 Determine the earliest stabilization index and sharp versions of the sufficient threshold $S > 2M + |c|$ of Corollary 3.4.

[Merge note, batch 73O1.] Item 43.12 is partly answered by source 50: its Theorem 3.2 replaces the threshold by the conditions $M < b \leq 2S - M$, $2S - b \notin H + H$, whose collision condition is sharp in the intake's random tests (see the note after Corollary 3.4). The earliest stabilization index for general (m, r) remains open.

Connections to broader dissociated-set theory (item 43.14 overlaps Question 5).

- 43.13 Dutta proved eventual exact doubling for the unrestricted greedy algorithm [5]. Find a common theorem that treats unrestricted doubling and congruence-restricted periodic corrections as two instances of one dynamical principle.

- 43.14 Extend the constrained greedy analysis to $D[g]$ sets, where each subset sum may have at most g representations, and to D_k sets with coefficients in $\{-k, \dots, k\}$.
- 43.15 Compare the asymptotic constants produced by residue restrictions with extremal constructions in the Erdős distinct-subset-sums problem [1, 4, 11].
- 43.16 Determine exact representation multiplicities inside the nearly saturated difference intervals. The support is simple, but the number of signed representations of each interior point may contain additional structure.

Asymptotic inversion and transseries (overlaps Question 8).

- 43.17 Develop a general inversion calculus for sequences of the form $\alpha\lambda^n + p(n)$ with periodic or automatic $p(n)$, retaining both exact staircase effects and smooth branch expansions.
- 43.18 For more complicated defect automata, identify when inverse expansions involve several exponential scales, logarithms, or genuine transseries rather than convergent $1/x$ series.
- 43.19 Formalize a theorem that transports error bounds from a forward exponential-polynomial approximation to residue-resolved inverse branches.

Formal verification and OEIS infrastructure (overlaps Questions 9 and 10).

- 43.20 Formalize the propagation theorem once in Lean and expose it as a reusable API for future OEIS certificates.
- 43.21 Build a checker that accepts a proposed defect set, base prefix, and periodic forcing table, and returns a machine-checkable recurrence certificate.
- 43.22 Search OEIS systematically for greedy subset-sum sequences whose conjectural recurrences can be resolved by this certificate pattern.

13 Conclusion

The two conjectured OEIS recurrences are manifestations of a stronger and simpler invariant. In shifted signed-sum coordinates, the reachable set is a complete interval except for a finite reflected fringe. Adding the next greedy term creates three translated copies. Once the copies overlap without a high-hole/low-hole collision, the same fringe reproduces itself. The arithmetic progression then reads that fringe through a finite residue table, forcing a periodic correction to doubling and hence a rational recurrence.

For A139217 the invariant has one fringe element, $H = \{3\}$; for A139218 it has four, $H = \{1, 3, 6, 11\}$. This proves the complete signed supports, the periodic deviations, the recurrences, closed forms, generating functions, asymptotics, and inverses. It also identifies and repairs the current A139217 recurrence range; for A139218 it proves the conjecture exactly as recorded. The same invariant yields exact counting functions, cross-sequence identities, and convergent inverse expansions (source 43). The broader finite-fringe method is likely to be more valuable than either isolated formula, because it reduces an infinite greedy problem to one exact seed computation and one finite collision check. More broadly, the results suggest that congruence-restricted greedy dissociated sets may often be governed by finite boundary automata; establishing that classification would convert a wide class of experimentally observed OEIS recurrences into certifiable theorems (source 43).

A Seed support certificates

A.1 A139217

For $A = \{1, 4, 7\}$, every integer in $[0, 12]$ except 3 has a ternary-digit representation:

x	representation	x	representation
0	0	1	1
2	$2 \cdot 1$	4	4
5	$4 + 1$	6	$4 + 2 \cdot 1$
7	7	8	$7 + 1$
9	$7 + 2 \cdot 1$	10	$2 \cdot 4 + 2 \cdot 1$
11	$7 + 4$	12	$7 + 4 + 1$

The value 3 is impossible because the contributions from 1 total at most 2, while the next available summand is 4. Reflection $x \mapsto 24 - x$ then proves (4.1). The signed form (4.3) was proved explicitly in Section 4 (source 43).

A.2 A139218

For $A = \{2, 5, 8\}$, the represented values in $[0, 15]$ are certified by

x	representation	x	representation
0	0	2	2
4	$2 \cdot 2$	5	5
7	$5 + 2$	8	8
9	$5 + 2 \cdot 2$	10	$2 \cdot 5$
12	$8 + 2 \cdot 2$	13	$8 + 5$
14	$2 \cdot 5 + 2 \cdot 2$	15	$8 + 5 + 2$

A direct check of the four remaining values gives the lower fringe $\{1, 3, 6, 11\}$; reflection $x \mapsto 30 - x$ gives the four upper holes. This proves (5.1).

A.3 Successive support addition (source 43)

Source 43 summarizes its hand certificate for A139218 by successive support addition. Starting from

$$D_0 = \{0\}, \quad D_k = D_{k-1} + \{-a_k, 0, a_k\},$$

with $(a_1, a_2, a_3, a_4) = (2, 5, 8, 14)$, one obtains

$$\begin{aligned} D_3 &= [-15, 15] \setminus \{\pm 4, \pm 9, \pm 12, \pm 14\}, \\ D_4 &= [-29, 29] \setminus \{\pm 18, \pm 23, \pm 26, \pm 28\}. \end{aligned}$$

The verification script `code/43-boundary-verify.py` evaluates the same equalities directly. No unproved numerical assumption enters the induction.

B Derivation of the rational generating functions

Suppose $a(n) = a(n-1) + a(n-2) + 2a(n-3)$ for $n \geq N$. With $A(x) = \sum_{n \geq 1} a(n)x^n$,

$$(1 - x - x^2 - 2x^3)A(x)$$

automatically has degree at most $N - 1$.

For A139217, using $a(1), \dots, a(5) = 1, 4, 7, 13, 22$ gives residuals

$$1, 3, 2, 0, -6,$$

which yields (4.8). For A139218, using $a(1), \dots, a(6) = 2, 5, 8, 14, 23, 41$ gives residuals

$$2, 3, 1, -3, -9, -12,$$

which yields (5.10).

Conversely, coefficient extraction from either rational function gives the listed initial values and the common recurrence after the numerator degree. Thus the generating functions are equivalent to the recurrence plus its transient data. (Source 43 argues the same way: multiply each power series by $1 - x - x^2 - 2x^3$; all coefficients vanish once the recurrence range begins, and expanding the two rational functions reproduces the OEIS prefixes.)

C Audit checklist

The logical dependencies of the report are intentionally short.

- (1) Lemma 2.1 turns greedy admissibility into nonmembership in a signed difference set.
- (2) Lemma 2.3 shifts the signed set to a nonnegative ternary support.
- (3) The exact seed computations identify H_{217} and H_{218} .
- (4) Theorem 3.2 preserves each fringe.
- (5) Proposition 3.5 converts the fringe into a finite residue table.
- (6) Theorem 3.6 converts periodic corrections into the recurrence.
- (7) Elementary induction gives the closed forms, generating functions, and inverse formulas.

No asymptotic assumption enters the proof of either OEIS result.

D Tables of terms, partial sums, and defects (source 43)

Table 1: Initial data for A139217. From $n = 3$ onward, the last column is the complete list of positive boundary defects.

n	u_n	U_n	positive missing value
1	1	1	—
2	4	5	—
3	7	12	9
4	13	25	22
5	22	47	44
6	49	96	93
7	97	193	190
8	190	383	380
9	385	768	765

n	u_n	U_n	positive missing value
10	769	1537	1534
11	1534	3071	3068
12	3073	6144	6141
13	6145	12289	12286
14	12286	24575	24572

Table 2: Initial data for A139218. From $n = 3$ onward, the last column is the complete list of positive boundary defects, in increasing order.

n	v_n	V_n	positive missing values
1	2	2	–
2	5	7	–
3	8	15	4, 9, 12, 14
4	14	29	18, 23, 26, 28
5	23	52	41, 46, 49, 51
6	41	93	82, 87, 90, 92
7	92	185	174, 179, 182, 184
8	179	364	353, 358, 361, 363
9	353	717	706, 711, 714, 716
10	716	1433	1422, 1427, 1430, 1432
11	1427	2860	2849, 2854, 2857, 2859
12	2849	5709	5698, 5703, 5706, 5708
13	5708	11417	11406, 11411, 11414, 11416
14	11411	22828	22817, 22822, 22825, 22827

Thus the u sequence has the single positive defect $U_n - 3$, whereas the v sequence has the four positive defects $V_n - 11, V_n - 6, V_n - 3, V_n - 1$.

[Merge note, batch 73O1.] The intake recomputed both tables from the greedy definition (brute-force signed difference sets); every entry agrees.

E Provenance and merge record

Source	Archive (arrival commit)	Main file, PDF	Pin	Author line
50 (base)	<code>finite_fringe_oeis_A139217_A139218.zip</code> (3a9518c52)	<code>finite_fringe_dissociated_sequences.tex</code> , 20 pp.	<code>c79d64038</code>	ProveIt research report (AI-assisted), prepared for Vladimir Reshetnikov with OpenAI
43	<code>ProveIt_OEIS_A139217_A139218.zip</code> (bdf1a1a73)	<code>article.tex</code> , 24 pp.	none	Research draft prepared for Vladimir Reshetnikov and the ProveIt project; computational assistant: OpenAI GPT-5.6 Sol Pro

Both are manuscripts of batch 73 (cluster O1) of ProveIt’s incoming-reports intake, numbered 50 and 43 there; the placement commit `9df4ba51a` staged their files and deleted the archives,

which survive in their arrival commits. Source 43 was written first (its archive members are timed 17:23 to 17:30 UTC on 1 October 2026; source 50's 18:33 to 18:39). Source 50's pin `c79d64038` already contains source 43's archive, but source 50 never cites it, and the longest passage the two texts share is 17 words of `cases` boilerplate or bibliography: they are independent.

What each contributed. Source 50: the base text, in its order, and everything in it. Source 43: the second proofs, the threshold form, the signed certificates, the partial-sum forcing principle, the cross-identities, the defect counts, the forward and spectral forms, the remainder bound and explicit inverse coefficients, the counting proofs and asymptotics, the second verifier, its OEIS wording, roadmap and questions, and the tables.

Where the merge chose.

- Base: source 50, the more general manuscript; its propagation theorem contains source 43's.
- Notation: source 50's throughout; source 43's symbols renamed as in Section 1.6, including the sign change $d_n \mapsto -c_n$, $e_n \mapsto -c_n$.
- The A139218 counting function uses source 43's residue-zero term, which is defined for every real x (merge note in Section 7.4).
- Source 43's “sharp” wording for the A139217 recurrence range and its $n \geq 4$ ranges for V_n and $14V_n - 13U_n$ are kept with merge notes giving the exact ranges.
- Source 50's sentence on the A290268 report is kept with a merge note. Its hard-coded “Section 6” became a cross-reference. Its uncited bibliography items (the two OEIS entries and the repository) are now cited.
- Source 43's title page, abstract, executive-summary box, package manifest and `listings` styling are not reproduced; their content is in Sections 1 and 9 and in the README. Its verbatim code excerpt is printed with `verbatim`.
- The two bibliographies are merged.

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- [9] V. Reshetnikov, *ProveIt*, GitHub repository, inspected at commit `c79d640385588fda4b9fdb0cab9fce44790fd100`, 1 October 2026 (source 50; source 43 cites the repository without a commit). <https://github.com/VladimirReshetnikov/ProveIt>
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